Review retrospective: *the harmonic monomial `GibbsObservable` witnesses — a clean pass, a dead open, a dead import, and a docstring that lied about its own file*

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Abstract

A soundness review of `Laplace/OneD/HarmonicGibbsObservableMonomials.lean`: the `Threepoint.GibbsObservable` witnesses for the harmonic Gibbs measure with monomial observables $x \mapsto x^k$ — the harmonic analogue of the anharmonic observable file, and the substrate that makes the harmonic cross-susceptibility capstone unconditional on the x, x^2, x^3 observables. Result: **a clean fidelity pass with four edits** — a dead `open`, a dead import, and a stale module docstring that described a file two-thirds smaller than the one it sat atop.

1 Setting

The harmonic counterpart of AnharmonicGibbsObservableMonomials: it builds the `Threepoint.GibbsObservable` data for $L = (\lambda/2)x^2$, perturbation $A = \mathrm{id}$, and every monomial observable $x \mapsto x^k$, by *dominated differentiation under the integral sign.* Its $k = 1, 2, 3$ specialisations are exactly the witnesses the harmonic cross-susc derivative capstone consumes to discharge the `GibbsObservable` hypotheses for x, x^2, x^3.

2 What was reviewed

The public surface: the $h = 0$ numerator reduction `harmonic_perturbed_numerator_zero_eq` (and its monomial form `..._pow`); the analytic ladder (`abs_mul_le_quarter_lambda_sq_add`, a Young-type absorption bound; `dominator_integrable_pow`; `harmonic_perturbed_integrand_pow_bound`; `harmonic_perturbed_integrand_pow_hasDerivAt`); the headline `Threepoint.harmonic_id_gibbsObservable_pow`; and the `..._id` / `..._mul_self` / `..._mul_mul_self` wrappers for x, x^2, x^3. The informal corpus was the file's own docstrings plus the sibling anharmonic file and the cross-susc consumer.

3 Fidelity / soundness findings

Sound; no defect in the mathematics. The reviewer confirmed each layer: the $h = 0$ identity is an extensional simplification of the integrand; the Young bound $|hx| \le (\lambda/4)x^2 + h^2/\lambda$ correctly absorbs the linear perturbation into the quadratic decay $e^{-(t\lambda/4)x^2}$; the base bound is correctly reused at index $k + 1$ and scaled by t to dominate the *derivative* integrand fed to hasDerivAt_integral_of_dominated_loc_of_deriv_le; and the x, x^2, x^3 wrappers are clean transports of the generic-k theorem along function equalities. "I found no proof-fidelity problem in the mathematics/proof structure of the theorems themselves."

One real fidelity issue — the docstring described the wrong file. The module header read "(in progress)"; the Status section claimed only the $h = 0$ primitive was provided and that the headline harmonic_id_gibbsObservable_pow "will land in a follow-up session." All false: the file already contained the full dominated-differentiation theorem and all three wrappers. The docstring was a snapshot of an *earlier* tide session, never refreshed when the headline landed. The fix rewrites the Status section to record what is actually present, and drops the "in progress" header. Two further doc drifts were corrected in the same edit: the strategy text named integrable_abs_pow_mul_exp_neg_mul_sq "already in the seabed (Laplace/OneD/IntegralRemainder.lean)", whereas the code uses Mathlib's integrable_rpow_mul_exp_neg_mul_sq from GaussianIntegral — both the *name* and the *home* were wrong.

4 Simplifications applied

A dead open and a dead import. open Set carried nothing (no bare Ioi/Iio/Icc/univ); open MeasureTheory stays (bare $\int$, Integrable, volume). And import Laplace.OneD.IntegralRemainder is dead: a full lake build (all consumers, 8306 jobs) is green without it. Signatures byte-identical, no warnings. The reviewer affirmed all four edits.

5 Disagreements and open questions

None substantive. The reviewer initially cautioned against marking the IntegralRemainder import dead "from source inspection," since the proofs textually reference integrable_rpow_mul_exp_neg_mul_sq — but explicitly allowed that "if a build later shows that theorem is re-exported transitively, removal may compile." It did: the lemma is a *Mathlib* lemma (GaussianIntegral, line 109), reachable through the retained HarmonicGibbsRegularity chain, not a denizen of IntegralRemainder at all. The reviewer's proof-golf suggestions (collapse the affine-derivative mini-block, inline two one-use rfl aliases) were declined as internal-shape-only.

6 Lessons

- **A textually-referenced lemma can still make its import dead — because the lemma lives upstream in Mathlib.** The proofs name integrable_rpow_mul_exp_neg_mul_sq, and the docstring even pointed at IntegralRemainder.lean as its home; but the lemma is Mathlib's, reachable via another retained import. "This symbol is used" does not imply "this import is load-bearing." The full build is the only arbiter — exactly mirroring the WeightedBridge/GaussianIntegral lesson on the qd seabed (a lemma's *name* need not match the file that exports it).
- **A module docstring is the doc site most prone to silent staleness.** It is written at the *start* of a tide, before the headline exists, and nothing forces a refresh when the headline lands. Here it underclaimed the file by two-thirds. When a file's header says "in progress" / "will land in a follow-up," check it against the actual declaration list before trusting it — and when fixing, prefer "provides ..." over "is complete" so it does not re-stale on the next addition.

7 Follow-ups

- The remaining never-reviewed laplace `OneD` files: `HarmonicAffineKappa3`, `HarmonicPerturbedMoments`, `QuarticBoundedPrior`, `QuarticBoundedPriorTestFn`, and the large `IntegralRemainder` itself.